Demand-Driven Manufacturing Optimization

A Big Data Analysis of Amazon E-Commerce Sales Using PySpark

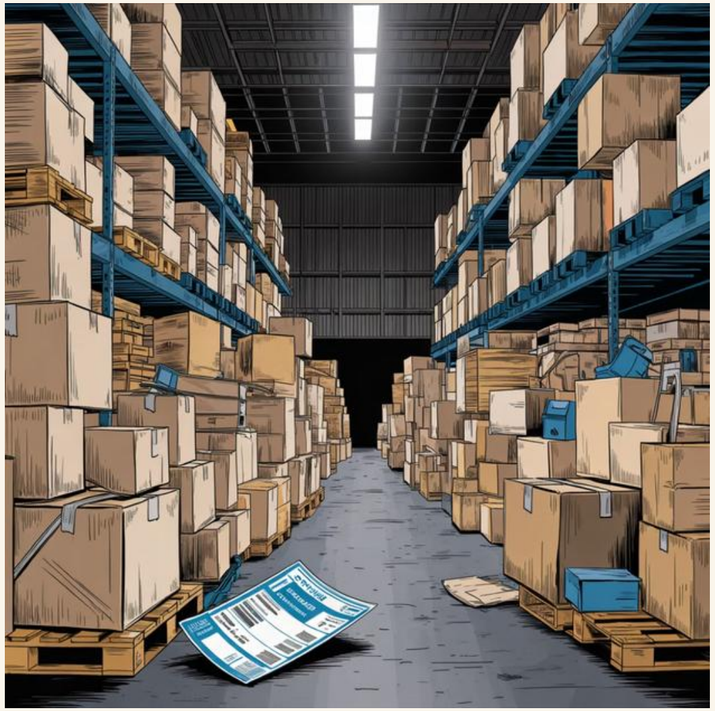
BIG DATA ANALYSIS

PYSPARK

By,

Lokeshwari.A

The Manufacturing Gap Problem



Overproduction Risk

Manufacturers routinely overproduce low-demand products — tying up capital, consuming raw materials, and generating avoidable waste and financial loss.

Underproduction Risk

High-demand products are underproduced — leaving revenue on the table and disappointing customers when stock runs out.

The Solution

Leverage **real Amazon sales data** to make smarter, data-backed production decisions before inventory decisions go wrong.

Dataset Overview

This analysis is grounded in one of the largest publicly available e-commerce datasets — the **Amazon Products Dataset 2023**, sourced from Kaggle and processed at scale using PySpark on Google Colab.

1.4M

Raw Products

Total products before cleaning

491K

Cleaned Records

After data quality filtering

248

Categories

Distinct product categories analyzed

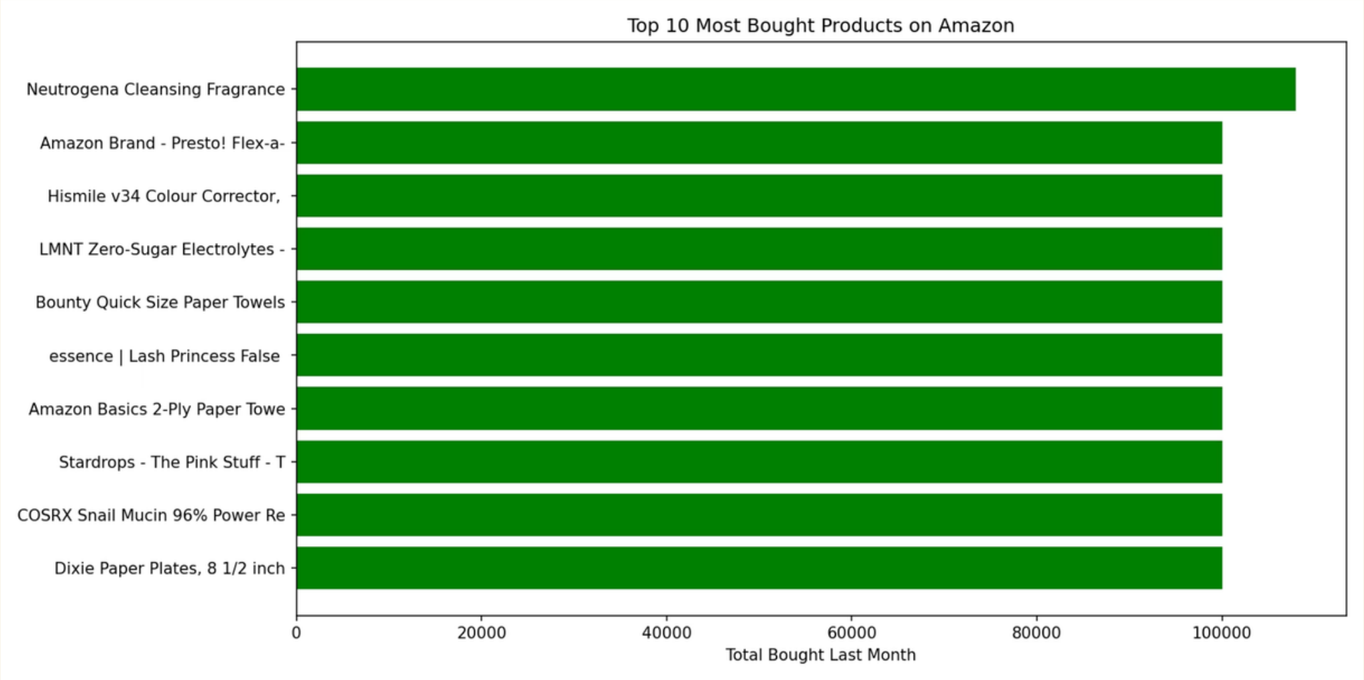
1

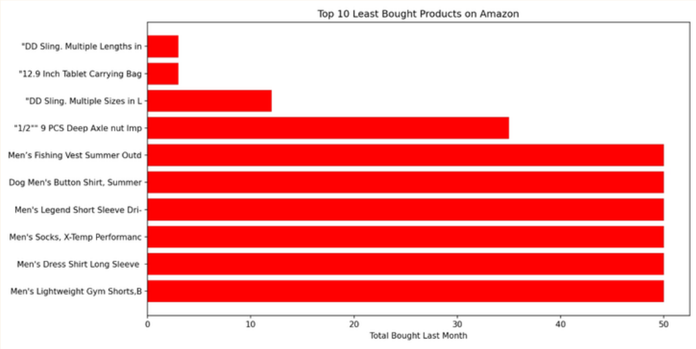
Key Signal

boughtInLastMonth — primary demand metric

What Sells — and What Doesn't

The spread between top and bottom performers is striking. Personal care and household essentials dominate volume, while niche accessories barely register.





🏆 Top Performers

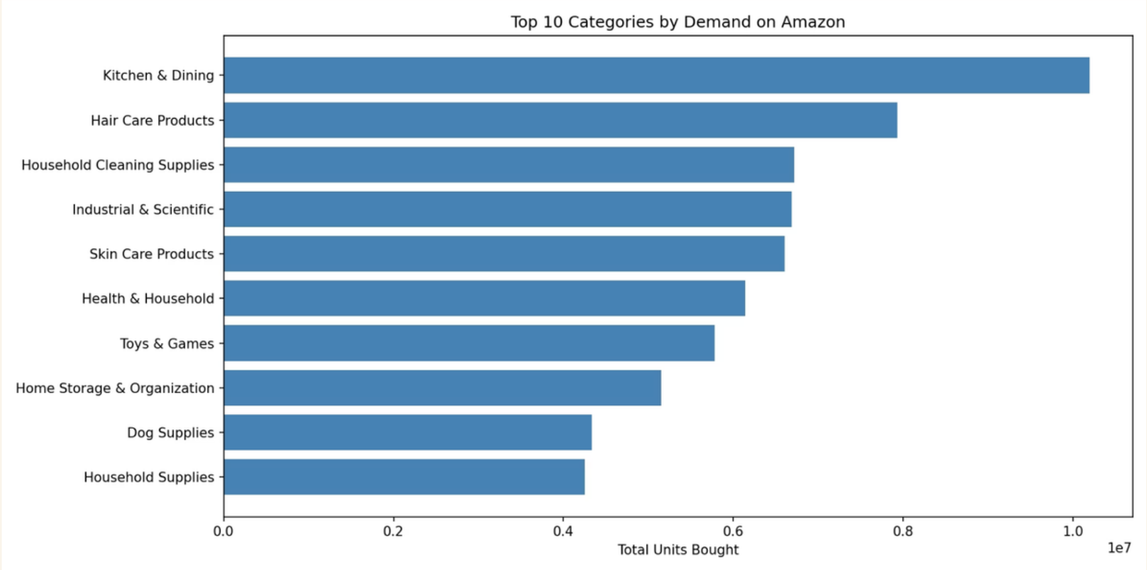
Neutrogena Face Wipes, Amazon Paper Towels & Hismile Teeth Whitening — each at **100,000 units/month**

⚠️ Lowest Performers

DD Sling (3 units), Tablet Carrying Bag & DD Sling 2"×6" (12 units each) — negligible market demand

Production Priority Score by Category

A composite **Production Priority Score** (0–100) was computed per category using normalized demand volume. Kitchen & Dining scores a perfect 100, anchoring the top of the ranking.



Kitchen & Dining — 100 🔥

Highest demand category by a wide margin

Hair Care — 77.82

Strong sustained consumer demand

Household Cleaning — 65.93

Essential goods driving consistent volume

Industrial & Scientific — 65.59

Broad SKU range with solid aggregate demand

Household Supplies — 41.71 ⬇️

Lowest-ranked category — reduce production priority

Pareto Analysis & Waste Estimation

The 80/20 Rule in Action

A Pareto analysis of the full dataset reveals a striking concentration of demand. Manufacturers can achieve **80% of total market coverage** by focusing on just **26% of their product catalog** — roughly 1 in every 4 SKUs.

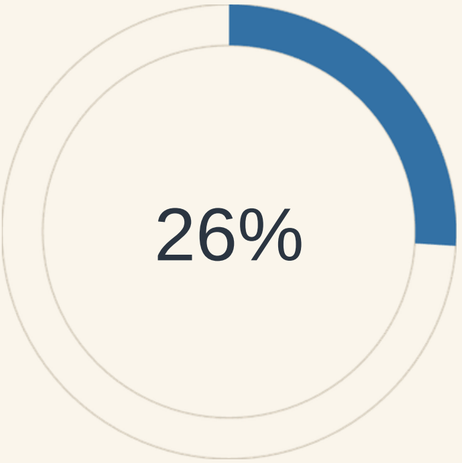
Simplifying the production portfolio doesn't mean losing market share — it means eliminating the noise.

Estimated Manufacturing Waste

Analyzing the **100 least-bought products** under conservative production assumptions reveals significant avoidable cost:

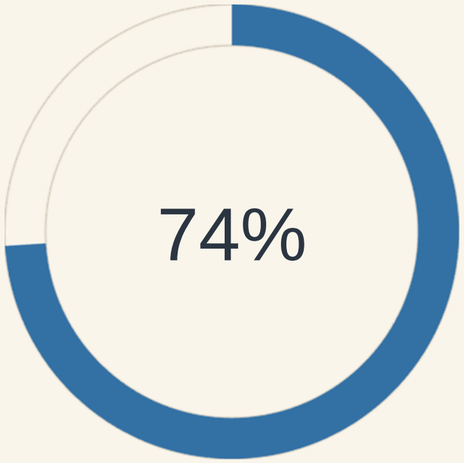
- 1,000 units produced per low-demand product
- Assumed cost: $5 per unit
- Estimated waste: **$475,735 (~5 Lakh USD)**

This loss is entirely preventable with data-driven production planning.



Products Needed

Drive 80% of total demand



Low-Priority SKUs

Candidates for production reduction

Waste Avoidable

$475K recoverable with data-driven decisions

Key Takeaways & Next Steps

Immediate Recommendations

- **Increase production** for Kitchen & Dining and Hair Care — consistently top-ranked categories
- **Reduce or halt** production of Household Supplies and DD Sling variants — demand signals are near zero
- **Reallocate $475,735** currently lost to low-demand overproduction toward high-priority SKUs
- **Concentrate portfolio** on the top 26% of products to capture 80% of market demand efficiently

Future Scope

Real-Time Prediction

Build a demand forecasting model using **PySpark MLlib** for dynamic production scheduling

Live Data Integration

Connect live Amazon sales feeds to enable **continuous, adaptive insights** rather than static snapshots